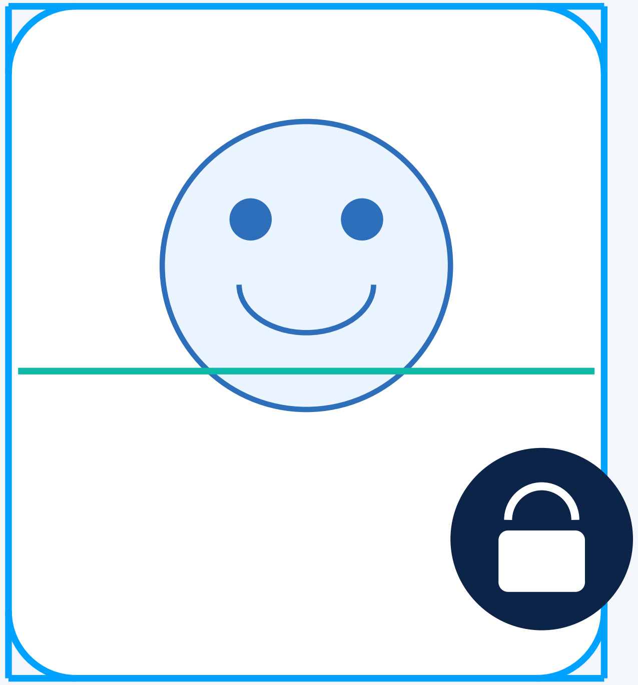
AI CAPSTONE PROJECT

AI-Powered Smart Lock System

Combining Facial Recognition and Real-Time Analytics for Optimal Home Security Management



FACE VERIFIED

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INTRODUCTION

- Smart home security systems are increasingly adopting AI and IoT technologies.
- Traditional keys, passwords, and RFID-based systems have limitations such as loss, theft, duplication, and unauthorized sharing.
- Facial recognition provides a contactless method for identifying authorized users.
- IoT enables communication between the recognition system and the physical lock.
- Real-time analytics can monitor access attempts and identify suspicious activities.
- This project combines **AI facial recognition, hardware-based access control, and a web-based analytics system**.

OBJECTIVES

1. Develop an AI-based facial recognition system for authentication.
2. Identify authorized and unauthorized users in real time.
3. Automate door locking/unlocking using ESP32 and SG90 servo.
4. Provide alerts for unauthorized access.
5. Record access events and timestamps.
6. Provide a web dashboard for real-time security analytics.

LITERATURE SURVEY

Author / Year	Focus	Key Contribution	Relevance
Khan et al., 2023	IoT Access Control	Reviews IoT access-control technologies and security requirements.	Supports secure IoT-based access-control design.
Mi et al., 2024	Privacy-Preserving Facial Recognition	Explores techniques for improving privacy in facial-recognition systems.	Supports secure handling of biometric authentication data.
Raj et al., 2025	Facial Recognition for Smart Homes	Presents a lightweight facial-recognition approach for real-time smart-home access control.	Supports real-time AI-based user authentication.
Zoha et al., 2025	IoT Smart Door Lock with Facial Recognition	Integrates facial recognition with an IoT-based smart door-lock system and security alerts.	Supports AI-based smart locking and access monitoring.
Thatikonda & Rani, 2025	AI and IoT Smart-Home Surveillance	Demonstrates AI-based monitoring with real-time security alerts.	Supports real-time security monitoring and alert generation.

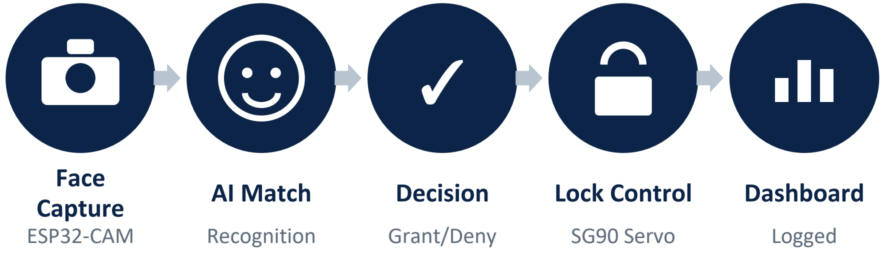
PROBLEM STATEMENT

- Traditional locks provide limited intelligent authentication.
- Keys/passwords can be lost, stolen, shared or misused.
- Unauthorized access may not be monitored continuously.
- Existing systems may lack detailed real-time analytics.
- Homeowners need centralized access history and security monitoring.
- An integrated AI + IoT + Web solution is required.

PROPOSED SOLUTION

- An AI-integrated smart lock platform is proposed to help users:
  - Authenticate users instantly through facial recognition.
  - Automatically lock or unlock the door based on the AI decision.
  - Alert instantly on unauthorized access attempts.
  - Log every access event with a precise timestamp.
  - Notify the admin dashboard in real time.
  - Display live security analytics for monitoring.

SYSTEM WORKFLOW



The workflow begins with a face scan captured by the ESP32-CAM. The AI module matches the face and issues a grant or deny decision, which drives the servo lock and instantly updates the security dashboard.

<u>Implementation:</u>

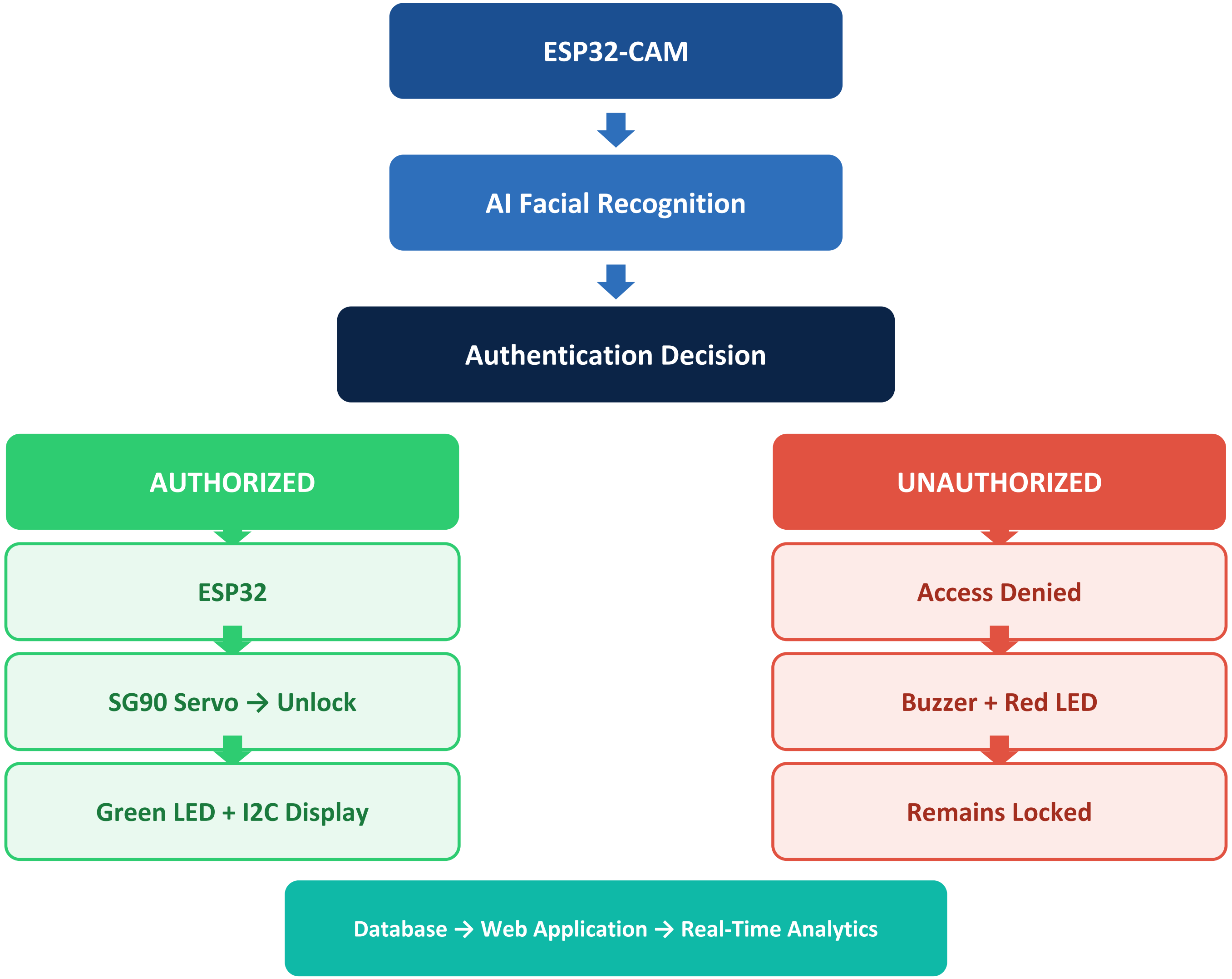
- Hardware: ESP32, ESP32-CAM, SG90 Servo
- AI: Python + OpenCV (Facial Recognition)
- Web: HTML, CSS, JavaScript, Flask, SQLite

COMPONENTS USED

Component	Purpose
ESP32-CAM	Captures face for facial recognition.
ESP32 DevKit	Main controller and communication.
SG90 Servo Motor	Lock/unlock mechanism.
Buzzer	Unauthorized access alert.
I2C Display	Displays system/access status.
Red LED	Locked / unauthorized indication.
Green LED	Authorized / unlocked indication.
Breadboard	Circuit prototyping.
Jumper Wires	Component connections.
5V Power Supply	Powers the circuit.

Engineer to Excel

SYSTEM ARCHITECTURE



MODULES OVERVIEW

Module 1

AI-Based Facial Recognition

- Face capture
- Face detection
- Feature extraction
- Face matching
- Authentication

Module 2

Smart Lock & Access Control

- ESP32
- Servo locking / unlocking
- I2C Display
- LEDs
- Buzzer

Module 3

Real-Time Security Analytics & Web App

- Access logging
- Database
- Dashboard
- Access history
- Security analytics
- Unauthorized-attempt monitoring

MODULE 1: AI-BASED FACIAL RECOGNITION

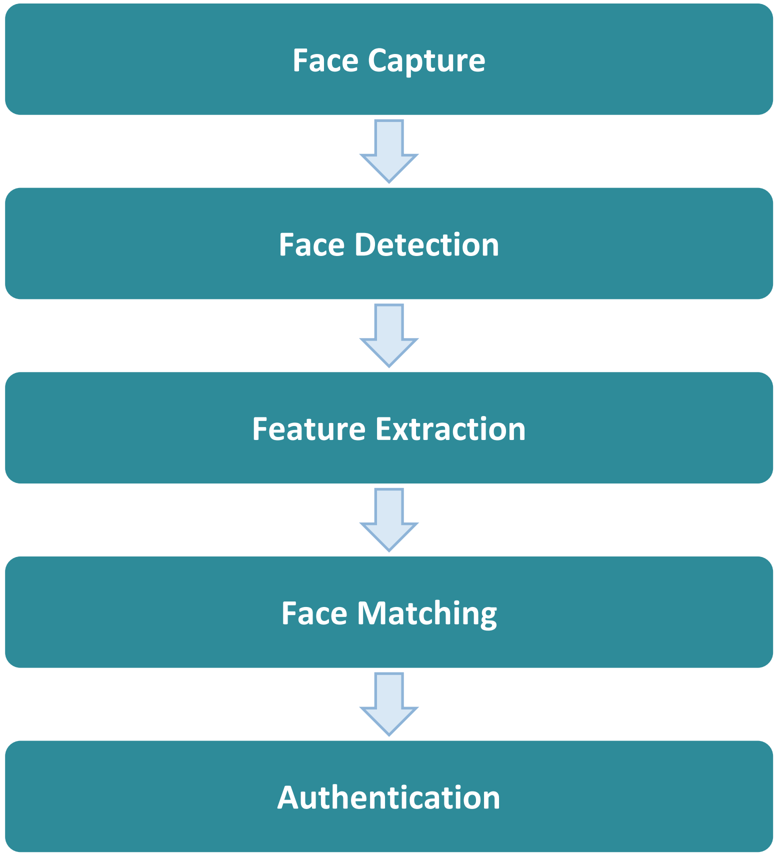
Purpose

- To authenticate users in real time by matching a captured face against registered facial data.
- To eliminate reliance on physical keys or passwords for entry.
- To ensure only verified individuals gain access to the property.

Key Functions

- Capture the user's face using the ESP32-CAM.
- Detect and align the facial region.
- Extract unique facial features.
- Match features against stored records.
- Confirm or reject the authentication request.

SYSTEM WORKFLOW



MODULE 2: SMART LOCK & ACCESS CONTROL

Purpose

- To convert the AI's authentication decision into a physical locking or unlocking action.
- To provide instant visual and audio feedback on every access attempt.
- To maintain a secure, fail-safe locked state by default.

Key Functions

- Receive the authentication decision from the AI module.
- Drive the SG90 servo to lock or unlock the door.
- Trigger the Green or Red LED indicator.
- Sound the buzzer on denied access.
- Update the access status on the I2C display.

SYSTEM WORKFLOW

Access Decision

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ESP32 Controller

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Servo Actuation

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LED & Buzzer Signal

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I2C Display Update

MODULE 3: REAL-TIME SECURITY ANALYTICS & WEB APP

Purpose

- To record every access event and present real-time security analytics.
- To give homeowners centralized visibility into who accessed the property and when.
- To support quick identification of unauthorized access patterns.

Key Functions

- Log every access attempt with a timestamp.
- Store records in the backend database.
- Render live dashboard statistics.
- Track unauthorized access attempts.
- Display access history and security trends.

SYSTEM WORKFLOW

Access Event

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Flask Backend

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SQLite Database

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Dashboard Rendering

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Real-Time Analytics

ADVANTAGES

Enhanced security through facial authentication

Contactless identification

Real-time access decisions

Unauthorized-access alerts

Centralized security analytics

Automatic access history

Web-based monitoring

Low-cost prototype

AI + IoT + Web integration

FUTURE SCOPE

- Multi-factor authentication
- Liveness detection
- Cloud monitoring
- Mobile application
- ML-based anomaly detection
- Edge AI
- Encrypted biometric storage
- Smart-home integration
- Voice security alerts

CONCLUSION

- ✓ Combines AI facial recognition, IoT hardware and web analytics.
- ✓ Provides automated contactless authentication.
- ✓ ESP32 + SG90 provide physical access control.
- ✓ LEDs, buzzer and I2C display provide immediate feedback.
- ✓ Web dashboard provides real-time monitoring and access history.
- ✓ Provides a low-cost prototype for intelligent home security.

REFERENCES

[1] A. Khan et al., “A Comprehensive Systematic Review of Access Control in IoT: Requirements, Technologies, and Evaluation Metrics,” IEEE Access, 2023.

[2] Y. Mi et al., “Privacy-Preserving Facial Recognition Using Trainable Feature Subtraction,” 2024.

[3] A. Raj, M. Marimuthu, and K. Lekshmi, “Smart Home Face Recognition: A Lightweight Approach for Real-Time Access Control,” 2025.

[4] M. S. Zoha et al., “The Development of IoT Based Smart Door Lock and Fire Alert System Using Facial Recognition,” 2025.

[5] S. Thatikonda and S. Sandhya Rani, “Smart Home Surveillance, Monitoring, and Automation with Real-Time Alerts via Raspberry Pi,” 2025.

THANK YOU

Questions & Answers